V.L Memorial Public School

475 Vijay Nagar

CHAPTER-6-QUESTIONS-CLASS6TH-SCIENCE

Class 06 - Science (Curiosity)

Section A

1. Which one of the following object is translucent? **[1]**

   a) cardboard
   b) cloud
   c) glass
   d) groundnut oil

2. Which of the following best describes classification? **[1]**

   a) The process of arranging items based on a common property
   b) Identifying objects by name
   c) Grouping objects by color
   d) Dividing objects by their size

3. Which of the following are translucent material? **[1]**

   a) air and clouds
   b) ground glass and air
   c) ground glass and clouds
   d) ground glass and aluminium sheet

4. What technique was developed around 4000 BCE for pottery production? **[1]**

   a) Hand crafting
   b) Wheel-turned pottery production
   c) Mould casting
   d) Metal coating

5. What does the ability to compress or scratch indicate about a material? **[1]**

   a) It describes the material's weight.
   b) It indicates transparency.
   c) It distinguishes between hardness and softness.
   d) It reflects the material's luster.

6. How would you classify an object that is made from one material? **[1]**

   a) Single-material object
   b) Composite
   c) Multi-material object
   d) Compound

7. A student observes a freshly cut piece of iron and notices that its surface shines. Which property is being identified? **[1]**

   a) Its lustrous appearance
   b) Its insoluble nature
   c) Its opaque appearance
   d) Its translucent appearance

8. What does the volume of matter represent? **[1]**

   a) Space occupied by matter
   b) Weight of matter present
   c) Amount of matter present
   d) Appearance of matter present

9. Which of the following would most likely not dissolve in water? **[1]**

a) Sugar
b) Lemon juice
c) Salt
d) Sand

10. A student scratches wood and stone with a metal key. The stone is harder to scratch. What can the student conclude? **[1]**

a) Stone is softer than wood
b) Wood is denser than stone
c) Stone is harder than wood
d) Wood is heavier than stone

11. How was pottery clay processed before shaping? **[1]**

a) Mixed with water
b) Burnt directly
c) Painted first
d) Selected, cleaned, sieved, and kneaded

12. Which one of the following is a transparent material? **[1]**

a) Air
b) Tissue paper
c) Butter paper
d) Oiled paper

13. Tables are made up of **[1]**

a) leather
b) both plastic and wood
c) plastic
d) wood

14. Which material is considered opaque? **[1]**

a) Frosted glass
b) Wood
c) Transparent plastic
d) Glass

15. What happens when sugar is mixed with water? **[1]**

a) The water changes color.
b) The sugar dissolves in the water.
c) The sugar evaporates.
d) The sugar remains undissolved, creating a layer.

16. Which cultural region's pottery is displayed at the National Museum, New Delhi? **[1]**

a) Greek
b) Sindhu-Sarasvati region
c) Mesopotamian
d) Egyptian

17. What is the primary purpose of classification? **[1]**

a) To arrange objects based on shared properties
b) To list objects alphabetically
c) To test the durability of objects
d) To maintain records accurately

18. **Assertion (A):** Materials can be grouped on the basis of whether they are transparent, translucent or opaque. **[1]**
Reason (R): A plastic bottle is transparent while ceramic is opaque material.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

19. **Assertion (A):** Mustard oil is soluble in water. **[1]**
Reason (R): Kerosene can mix with petrol to form a single layer.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

20. **Assertion (A):** Froasting glass door is neither fully transparent nor completely opaque. **[1]**
Reason (R): Translucent materials allow some light to pass through but not enough to see objects clearly.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

21. **Assertion (A):** We classify the objects into groups. **[1]**
Reason (R): The classification of objects into groups helps in understanding them.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Anything that occupies space and has mass is called matter. **[1]**
Reason (R): The space occupied by matter is its volume.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Metal is a hard, shiny, solid material. **[1]**
Reason (R): Copper and aluminium are metals.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Grouping materials based on their common properties aids in better understanding and utilization of these materials. **[1]**
Reason (R): The method of arranging the objects into groups is called classification, and classification helps to systematically organize knowledge.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. Why do we choose a particular material for preparing an object? **[1]**

26. A boy added 10 g of common salt to 250 mL of water. The salt disappeared. Is the salt still present in the water, or has it evaporated? **[1]**

27. When we pour a few drops of mustard oil into water, the oil floats on the surface of **[1]**

the water. Why?

28. Why is it important to classify objects based on common properties? **[1]**

29. Which can be compressed easily-a sponge or a lemon? **[1]**

30. Harshit was able to see cricket match that his friends were playing in the park through a window, although the window was closed. How? **[1]**

31. How can classification help in selecting materials for making a tumbler? **[1]**

Section B

32. Match the following: **[2]**

Object	Materials
(a) Chair	(i) Leather
(b) Book	(ii) Glass
(c) Tumbler	(iii) Paper
(d) Shoes	(iv) Wood

33. **Match the Following:** **[2]**

Column (A)	Column (B)
(a) Terracotta	(i) Baked clay
(b) Wheel-turned pottery	(ii) Sindhu–Sarasvatī Civilisation
(c) Decorative slips	(iii) Protective or decorative coats
(d) Storage jars	(iv) National Museum, New Delhi

34. Evaluate the challenges faced when classifying objects with overlapping properties. **[2]**

35. Consider a scenario where plastic is used to make different containers. How would you classify these containers beyond the material used? **[2]**

36. Create a method to classify materials that could be used to make a durable tumbler. **[2]**

37. Explore how the standard of leaving a space between the numerical value and unit affects scientific notation and clarity. **[2]**

38. Design an experiment to test which materials, out of a given set, can effectively hold water. **[2]**

39. How would you apply the principle of classification to organize everyday household items? **[2]**

40. Analyze how the concepts of classification and material selection impact sustainable product design. **[2]**

Section C

41. Evaluate the connection between the different materials used for pottery and their roles in the socio-economic structure of ancient civilizations. **[3]**

42. Provide an example of how the classification of materials can be applied in everyday life. **[3]**

43. Apply the principle of classification to organize the following items: plastic bottle, wooden spoon, and cotton cloth. Justify your classification. **[3]**

44. Analyze how the International System of Units (SI) facilitates consistency in scientific **[3]**

communication.

45. Identify the materials commonly exhibited at the National Museum in New Delhi and their historical relevance. **[3]**

46. Describe the various materials that objects are made of and discuss any one method of how clay is transformed into pottery. **[3]**

47. Evaluate the importance of understanding the properties of materials like transparency, hardness, and solubility in everyday life. **[3]**

Section D

48. **Read the following text carefully and answer the questions that follow:** **[4]**

When you press different objects or materials with your hands, some of them like stones, may be hard to compress, while others, like an eraser, can be easily compressed. Materials which can be compressed or scratched easily are soft, while other materials which are difficult to compress or scratch are hard. However, these properties are relative in nature. For example, rubber is harder than sponge but softer than iron.

a. Give an example of a material that is hard to compress. (1)

b. Which is harder: rubber or sponge? (1)

c. Explain the difference between hard and soft materials with examples. (2)

OR

Why is hardness considered a relative property? Explain with an example. (2)

49. **Read the following text carefully and answer the questions that follow:** **[4]**

Materials that typically have shiny surfaces are said to have a lustrous appearance. Such materials with lustre are usually metals. Examples of metals include iron, copper, zinc, aluminium, gold, etc. However, some metals may lose their lustre and start to look dull or non-lustrous due to the effect of air and moisture on them. As a result, we often notice the lustre only on their freshly cut surfaces. Non-lustrous materials are those that do not have a shiny surface. Some examples of non-lustrous materials are paper, wood, rubber, jute, etc.

a. Name two metals that have a lustrous appearance. (1)

b. Give an example of a non-lustrous material. (1)

c. Why do some metals lose their lustre over time? How can their lustre be restored? (2)

OR

Differentiate between lustrous and non-lustrous materials with examples. (2)

50. **Read the following passage carefully and answer the questions that follow:** **[4]**

Ghulan and Sheeta are thrilled to go to their new class after the summer break. They enter their classroom and start talking. “What have you brought to school today?” asks Sheeta. After some time, Madam Vidya, their science teacher, enters the class and starts interacting with the students to draw their attention to many things they use in their daily life. She asks, “How are they similar to or different from each other? What are their shapes and colours? How does it feel when you touch them? Are some of them heavier than others?” All things are made up of some materials like

paper, wood, cloth, glass, metal, plastic, clay, and so on. Any substance that is used to create an object is referred to as a **material**. The earliest pottery found in the Indian subcontinent dates back to thousands of years ago. During the Sindhu-Sarasvatī (Harappan) Civilisation, people developed techniques of wheel-turned pottery, applied decorative coats, painted designs, and baked the clay in kilns. Baked clay is called **terracotta**. Pots were used for cooking and for storing food, grains, oil, and ghee.

Questions:

a. What are the different materials mentioned in the passage that objects can be made from? **(1)**

b. What is 'terracotta' as described in the passage? **(1)**

c. Describe how the use of clay and pottery techniques evolved during the Sindhu-Sarasvatī Civilisation . **(2)**

OR

How does pottery reflect the creativity and skills of early people? **(2)**

Section E

51. How can understanding the properties of materials like transparency, solubility, and texture assist in practical applications? Provide examples. **[5]**

52. Discuss the significance of mass, volume and material properties in describing matter. **[5]**

53. If you drop a wooden block and a metal key in water, what will happen? Explain why. **[5]**

54. Why do we use different materials to make different objects? Explain with examples. **[5]**

55. If you are given an object, how will you identify whether it is made of metal or plastic? **[5]**

56. Describe how the introduction of wheel-turned pottery in the Sindhu-Sarasvatī Civilization impacted the efficiency and artistry of pottery production. **[5]**

57. Differentiate between: **[5]**

a. Opaque and transparent materials.

b. Soluble and insoluble.

c. Lustrous and non-lustrous.

d. Malleable and brittle.

Section F

58. **State whether the given statement is True or False:** **[9]**

(a) Glass is transparent. **[1]**

(b) Plastic can be used to make a variety of objects. **[1]**

(c) Grouping is done on the basis of similarities in things. **[1]**

(d) Mass is measured in units such as litre (L) and millilitre (mL). **[1]**

(e) Grouping refers to arranging objects based on dissimilar properties. **[1]**

(f) Classification of objects relies on identifying a shared characteristic. **[1]**

(g) The process of arranging objects into groups is known as classification. **[1]**

(h) Kilogram is abbreviated as kg with no space between the letters. **[1]**

(i) Rubber is a hard and brittle material. **[1]**

59. **Fill in the blanks:** **[9]**

(a) The materials that are required to make a tumbler should be capable of holding _____. **[1]**

(b) Kilogram is the unit of _____ in the International System of Units (SI). **[1]**

(c) Baked clay is commonly referred to as _____. **[1]**

(d) Freshly cut wood has a distinct appearance different from that of _____. **[1]**

(e) Grouping of things is done for _____. **[1]**

(f) The technique of wheel-turned pottery production began around _____ BCE. **[1]**

(g) When sugar is mixed in water, it _____ after a while. **[1]**

(h) The units to measure the volume of matter are liter (L) and _____. **[1]**

(i) A material used to create an object is referred to as _____. **[1]**